

Expérience sur les oscillations d'un ressort spiral en régime libre et en régime forcé

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Abstract

In this experiment we studied a very important physical effect by measuring the dependence of a quantity V on the quantity X for two different sample temperatures. Our experimental measurements confirmed the quadratic dependence $V = kX^2$ predicted by Someone's first law. The value of the parameter $k = 15.4 \pm 0.5$ s was extracted from the fit. This value is not consistent with the theoretically predicted $k_{theory} = 17.34$ s. We attribute this discrepancy to low efficiency of our V -detector.

Contents

1	Introduction	3
2	Description des installations	3
2.1	Expérience 1	3
2.2	Expérience 2	3
3	Théorie	3
3.1	Expérience 1	3
3.2	Expérience 2	3
4	Description des manipulations	3
5	Résultats	3
5.1	Expérience 1	3
5.2	Expérience 2	3
6	Analyse des résultats	4
7	Conclusion	4
8	Sources	4
9	Annexes	4
9.1	Annexe A	4
9.2	Annexe B	4
9.3	Annexe C	4

1 Introduction

The very important physical effect has applications to astronomy, nuclear physics, condensed matter, and more.

2 Description des installations

2.1 Expérience 1

Texte décrivant l'installation de l'expérience 1.

2.2 Expérience 2

Texte décrivant l'installation de l'expérience 2.

3 Théorie

3.1 Expérience 1

According to references [1–3] the dependence of interest is given by

$$u(\lambda, T) = \frac{8\pi hc\lambda^{-5}}{e^{hc/\lambda kT} - 1}, \quad (1)$$

3.2 Expérience 2

Complément théorique pour l'expérience 2.

4 Description des manipulations

Give a schematic of the experimental setup(s) used (figure 1).

Figure 1: Every figure MUST have a caption.

5 Résultats

5.1 Expérience 1

Insert tables/plots here.

5.2 Expérience 2

Insert tables/plots here.

6 Analyse des résultats

Here you explain uncertainties, systematic errors, graphical analysis, etc.

Figure 2: Every plot must have axes labeled.

7 Conclusion

Here you briefly summarize your findings.

8 Sources

References

- [1] A. C. Melissinos and J. Napolitano, *Experiments in Modern Physics*, (Academic Press, New York, 2003).
- [2] N. Cyr, M. Têtu, and M. Breton, IEEE Trans. Instrum. Meas. **42**, 640 (1993).
- [3] *Expected value*, available at http://en.wikipedia.org/wiki/Expected_value.

9 Annexes

9.1 Annexe A

Contenu de l'annexe A.

9.2 Annexe B

Contenu de l'annexe B.

9.3 Annexe C

Contenu de l'annexe C.